LUMAFLOW · REGULATORY AND STANDARDS REGISTER

LumaFlow regulatory and standards register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	Kit 1.8 upgrade: first register for the TRL 3 design (option A), with notes for option B

LumaFlow regulatory and standards register

Which standards and regulations would apply to LumaFlow as a product, where, and what is needed. Not legal advice; anything marked "to assess" needs a qualified reviewer before any product claim. LumaFlow is a research and educational prototype, not a certified water treatment device (PR-019), and no certification is pursued while the portfolio is capped at TRL 3. Method: STANDARDS section 23. Water-security repos in the portfolio share this research; each row was checked on the date in its source.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	NSF/ANSI 55, ultraviolet microbiological water treatment systems (Class A: 40 mJ/cm² at rated flow and 70 % UVT or the alarm set point, with a UV sensor and alarm; Class B: 16 mJ/cm²)	United States and Canada (voluntary, often required by health authorities)	Performance, sensor and alarm of UV point-of-use and point-of-entry systems	The dose target (PR-002) borrows the Class A dose level; PR-003 (the 70 %/cm condition) is not met, so a Class A claim is not possible as designed	Design lead	Certification needed later; Class A not met (PR-003)
G2	NSF/ANSI 61, drinking water system components, health effects	United States and Canada	Contaminants leached from materials in contact with drinking water	Every wetted part: 316, PTFE, expanded PTFE, fused silica, EPDM O-rings, acetal, push-fit fittings	Design lead with a certification body	To assess (choose certified grades at part selection)
G3	NSF/ANSI 372, lead content	United States	Weighted average lead content of wetted surfaces	Stainless and polymer wetted parts; push-fit fittings and stem adaptors must be lead-free grades	Design lead	To assess
G4	21 CFR 177.1550 (perfluorocarbon resins), 177.2470 (polyoxymethylene copolymer), 177.2600 (rubber articles)	United States	Food-contact materials	PTFE liner, top disc, washers and expanded PTFE gasket; acetal upper cap; EPDM O-rings	Design lead	Applies: specify compliant grades (PR-011)
G5	IEC 60335-2-109, safety of UV radiation water treatment appliances	International (adopted regionally, for example as EN 60335-2-109)	Electrical and UV radiation safety of household UV water treatment appliances	Applies to LumaFlow as a product: enclosure, interlocks (PR-012), marking	Electrical safety reviewer	Certification needed later; to assess with a test laboratory
G6	IEC 62471, photobiological safety of lamps and lamp systems	International	Risk groups for optical radiation, including UV-C	The LEDs are fully enclosed in use; applies to service with the head removed (warning labels, interlock)	Electrical safety reviewer	To assess
G7	IEC 62368-1 (or the regional equivalent) for the 24 V adapter	International	Safety of power supplies	Met by buying a certified adapter (PR-013)	Design lead	Met by design (bought, certified)
G8		United States; European Union		The microcontroller and switching LED driver are	Electronics lead	

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
	US FCC Part 15 and EU EMC Directive 2014/30/EU		Emissions from the controller and LED driver	unintentional radiators		To assess (product stage)
G9	EU RoHS Directive 2011/65/EU and WEEE Directive 2012/19/EU	European Union	Hazardous substances in and take-back of electrical equipment	LEDs, boards, adapter	Electronics lead	To assess (product stage)
G10	Minamata Convention on Mercury	International	Phase-out of mercury-added products, including lamps	LumaFlow uses no mercury (PR-006); the convention is part of the case for LED disinfection	Project lead	Met by design
G11	EU Drinking Water Directive (EU) 2020/2184, Article 11 (materials in contact with water)	European Union	Hygiene requirements for materials and products in contact with drinking water	Wetted materials if sold in the EU	Design lead	To assess
G12	WHO International Scheme to Evaluate Household Water Treatment Technologies	International	Performance classes by log reduction of bacteria, viruses and protozoa	A route to an independent performance rating; LumaFlow's virus reduction is small (STM S5)	Project lead with a partner laboratory	To assess (TRL 5 or later)
G13	US EPA Ground Water Rule and state well-water guidance	United States	Private wells are outside the Safe Drinking Water Act; guidance only	Context for the first user (private well household)	Project lead	Does not apply to the device
G14	Plumbing codes for point-of-use devices and backflow (for example the Uniform and International Plumbing Codes)	United States (local adoption)	Connection to the cold line, backflow, accessible shutoff	The installation (tee, pressure-limiting valve, pre-filter, PR-017)	Installer	To assess at installation

Status: to assess, applies, does not apply, met by design, certification needed later.

Option B (hollow-fibre ultrafilter), if chosen, would fall under NSF/ANSI 53 (health effects of filtration, including cyst reduction) or NSF/ANSI P231 or NSF/ANSI 244 for microbiological claims, in place of G1, G5 and G6; to assess.

Sources

- NSF/ANSI 55 Class A and B summary: Fresh Water Systems, "NSF Class A and Class B," https://www.freshwatersystems.com/blogs/blog/nsf-class-a-and-class-b, and ANSI blog, "NSF/ANSI 55-2024," https://blog.ansi.org/ansi/nsf-ansi-55-2024-ultraviolet-uv-water-treatment/, both cited in LMF-PRB-001 (checked 2026-09-25).
- NSF/ANSI 61 and 372 scope: UL Solutions, "Health effects and lead content of drinking water products," https://www.ul.com/services/health-effects-and-lead-content-drinking-water-products, accessed 2026-10-25.
- 21 CFR 177.1550 and 177.2470: https://www.law.cornell.edu/cfr/text/21/177.1550 and https://www.law.cornell.edu/cfr/text/21/177.2470, accessed 2026-10-25.
- IEC 60335-2-109:2023 listing (household UV water treatment appliances): Standards Council of Canada, https://scc-ccn.ca/standardsdb/standards/2053444, accessed 2026-10-25.
- Minamata Convention on Mercury, "Mercury-containing lamps phased out," https://minamataconvention.org/en/news/mercury-containing-lamps-phased-out-minamata-convention-offices-geneva, cited in LMF-PRB-001.
- WHO International Scheme to Evaluate Household Water Treatment Technologies, https://www.who.int/tools/international-scheme-to-evaluate-household-water-treatment-technologies, cited in LMF-PRB-001.

- Rows G6 to G9, G11, G14 and the NSF standards named for option B are listed from the standards' own titles; their applicability is "to assess" and needs a test laboratory or regulatory consultant.